Sheikh Mumin Murtaza

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.91
Delhi Public School Srinagar	Class XII (CBSE)	2024	94.20%
Delhi Public School Srinagar	Class X (CBSE)	2022	93.80%

PROJECTS

Expected Goals (xG) Prediction System | *ML Model*

Python, Scikit-learn, XGBoost, Streamlit

- Built an end-to-end xG prediction system trained on **3,000+ StatsBomb** shot events from professional matches.
- Engineered spatial and contextual shot features, training Logistic Regression and XGBoost models to **0.80 ROC-AUC**.
- Enforced monotonic constraints in **XGBoost** to eliminate unrealistic probability extrapolation and improve reliability.
- Deployed an interactive Streamlit app with shot maps, player profiles, and **live xG prediction** on new inputs.

Image Compressor | *Web Application*

Python, FastAPI, Pillow, HTML, CSS, JavaScript

- Built a **full-stack app** that compresses images to a user-defined file size or percentage, preserving visual quality.
- Engineered an adaptive compression algorithm using **binary search** over JPEG/WebP quality, with resolution scaling.
- Added rotation, horizontal/vertical flipping, and format conversion, all processed within the FastAPI backend.
- Deployed as a single **FastAPI service** serving both the REST API and frontend as one browser-based tool.

AttendX | *Face Recognition Attendance Platform*

Python, FaceNet, Faiss, FastAPI, Docker, HF spaces, Supabase

- Built real-time face attendance using **YuNet, FaceNet512,** and **FAISS** for sub-second similarity search across users.
- Prevented cross-tenant data leakage with isolated per-organization FAISS indexes and **JWT-based authentication**.
- Deployed AttendX with FastAPI, Docker, and Supabase, hosting the live platform on Hugging Face Spaces.
- Enabled **thread-safe attendance** marking using per-classroom RW locks on shared FAISS indexes for concurrency.

Mathematical Conjecture Solver | *Research on Numerical Methods*

Python, FastAPI, REST APIs

1. Built **REST APIs** with **FastAPI** to solve math conjectures using modular and reusable **Python** methods effectively.
2. Designed modular Python implementations for multiple algorithms, enabling efficient maintainable solutions.
3. Integrated backend services and conducted **API testing** to ensure reliable, scalable, consistent API release.
4. Benchmarked algorithm outputs against known mathematical results to validate **accuracy** and **performance.**

SKILLS

- **Programming Languages:** Python, C++, SQL
- **Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming, Backend Development
- **Machine Learning & AI:** Machine Learning, Regression, Clustering, Deep Learning, Neural Networks, Scikit-learn
- **Web Development & APIs:** FastAPI, RESTful APIs, Streamlit

- **Data & Developer Tools:** NumPy, Pandas, Matplotlib, Git, GitHub

RELEVANT COURSES

Mathematics & Computing:

- Artificial Intelligence, AI/ML Tools, Introduction to Programming
- Numerical Methods, Linear Algebra, Differential Equations, Applied Mathematics.

Other:

- Digital Fabrication, Basic Electrical Engineering.

POSITIONS OF RESPONSIBILITY

NSO Football Team Leader

- Led structured football training sessions, planning drills to improve player fitness, technical skills, and team coordination.
- Coordinated player participation, match scheduling, and on-field logistics for football events, ensuring smooth execution.

Football Community Volunteer

- Officiated football matches across five IITH tournaments, ensuring fair play and application of competition rules.
- Organized tournament logistics, including fixture scheduling, player auctions to support seamless execution.

EXTRACURRICULAR

Achievements & Sports

- Represented IIT Hyderabad at **Inter IIT Sports Meet** 2024 and 2025 (Football)
- **Gold Medal** -- Football, Diesta 2026
- **Bronze Medal** -- Football, Diesta 2025
- **Bronze Medal** -- 20 km Cyclathon
- **Bronze Medal** -- 5 km Marathon

Other Sports

- Represented the School Team in **CBSE Inter School Tournament** in Class XI (Football)
- Completed Beginner, Intermediate, and Advanced courses at **IISM Gulmarg.**
- **1st Place** in intermediate technique exam for skiing.